

Airbnb Price Prediction Report — Milan

Generated on 2026-08-25 by the automated pipeline (`make all`). Source: InsideAirbnb.

Executive summary

This report analyzes the **Milan** Airbnb market using 20,453 cleaned listings and 26 listing features. A machine-learning model was trained to predict nightly prices and then evaluated on a **holdout set** — listings it had never seen before.

The best model was **XGBoost**. On the holdout set it achieved:

- **Average error (MAE): \$54** — a typical prediction is off by about this much.
- **RMSE: \$107** — like MAE, but weighs large errors more heavily.
- **MAPE: 38%** — the average error as a share of the actual price.
- **R²: 0.50** — the share of price variation the model explains (0–1).
- **50%** of predictions were within +/-25% of the true price.

How to read this report

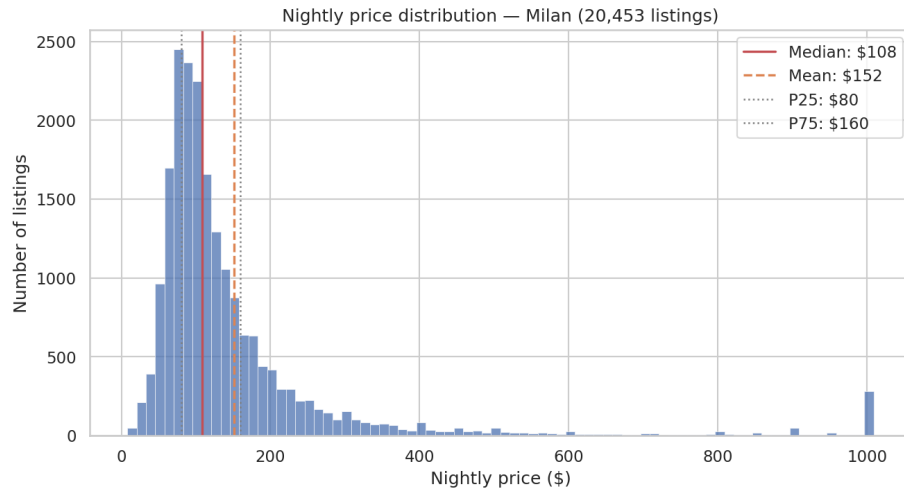
- **MAE (mean absolute error)** — the average difference between predicted and actual prices, in dollars. Lower is better; it's the easiest number to grasp ("off by about \$X on average").
- **RMSE (root mean squared error)** — similar to MAE but penalizes large mistakes more. If RMSE is much higher than MAE, a few big errors are inflating it.
- **MAPE (mean absolute percentage error)** — the same idea as MAE, but expressed as a percentage of the actual price, so it can be compared across cities.
- **R²** — how much of the variation in prices the model explains (0 = no better than guessing the average; 1 = perfect).
- **Holdout set** — the 10% of listings the model never saw while training. All accuracy numbers come from this set, so they are an honest estimate of real-world performance.
- **Top features** — the listing attributes the model leans on most when setting a price.

Data

- Source: InsideAirbnb (latest available snapshot)
- Listings after cleaning: **20,453** — median nightly price **\$108**, mean **\$152**
- Features used by the model: **26**
- Train / validation / holdout split: **14,317 / 3,681 / 2,455** listings (70/15/10, random seed 42)

Market overview

Price distribution



- Median nightly price: **\$108**; the middle 50% of listings fall between **\$80** and **\$160**. - The mean (**\$152**) sits well above the median — a minority of expensive listings pull the average up, so the **median is the better guide** to a typical price.

Price by room type

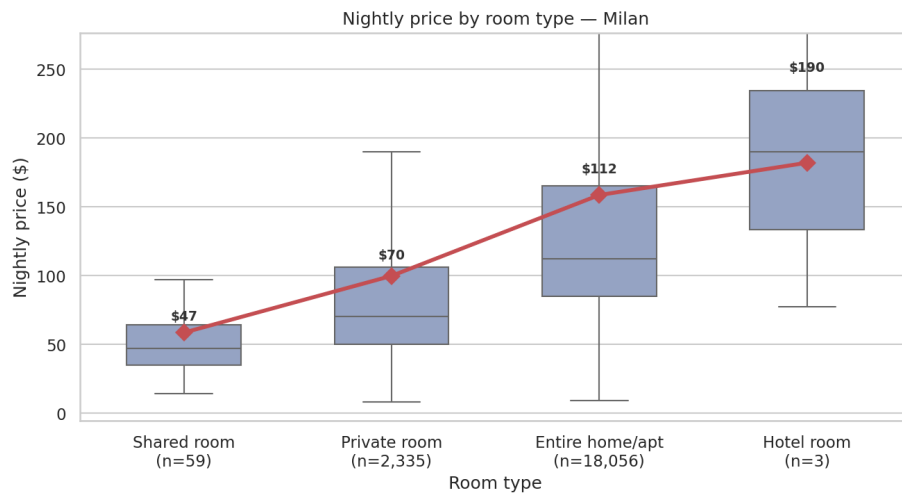


Figure 1: price_by_room_type

Room type	Listings	Median price	Mean price
Hotel room	3	\$190	\$182
Entire home/apt	18,056	\$112	\$159
Private room	2,335	\$70	\$100
Shared room	59	\$47	\$58

Price by number of bedrooms



Figure 2: price_by_bedrooms

Feature correlations

Model accuracy on the holdout set

Metrics for the chosen model (**XGBoost**) on 2,455 unseen listings:

Metric	Value	What it means
MAE	\$54	Average prediction is off by this much
RMSE	\$107	Errors, with big misses weighted more
MAPE	38%	Average error as a % of the true price
R ²	0.50	Share of price variation explained

Accuracy in practice: 21% of predictions were within +/-10% of the true price, 50% within +/-25%, and 76% within +/-50%.

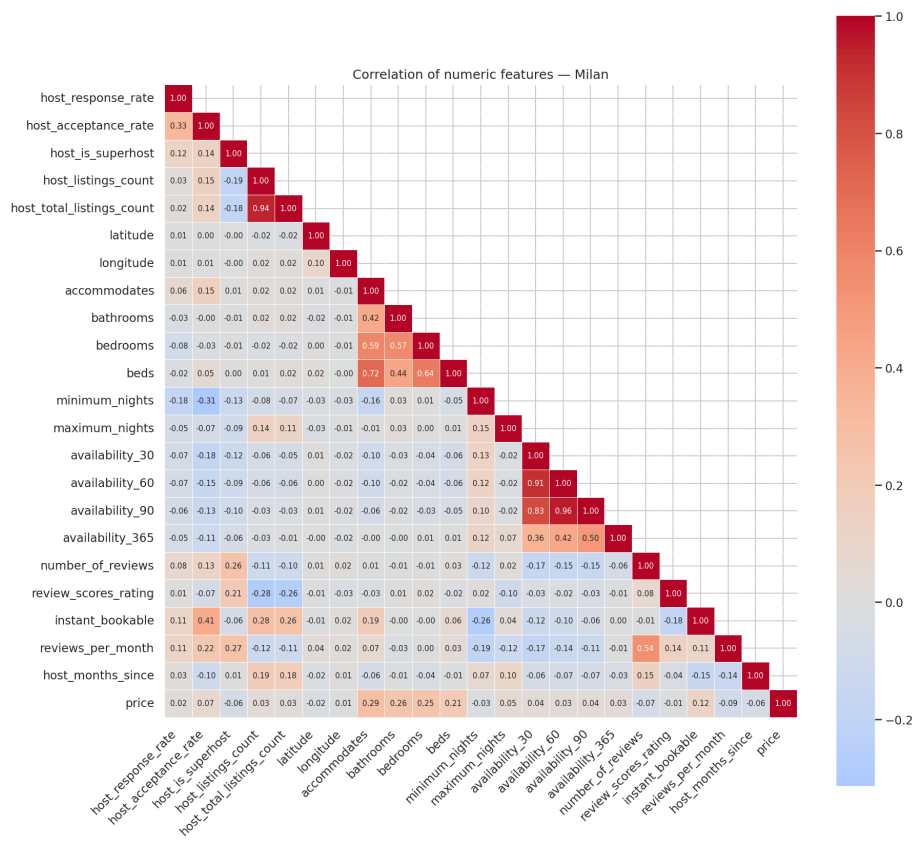
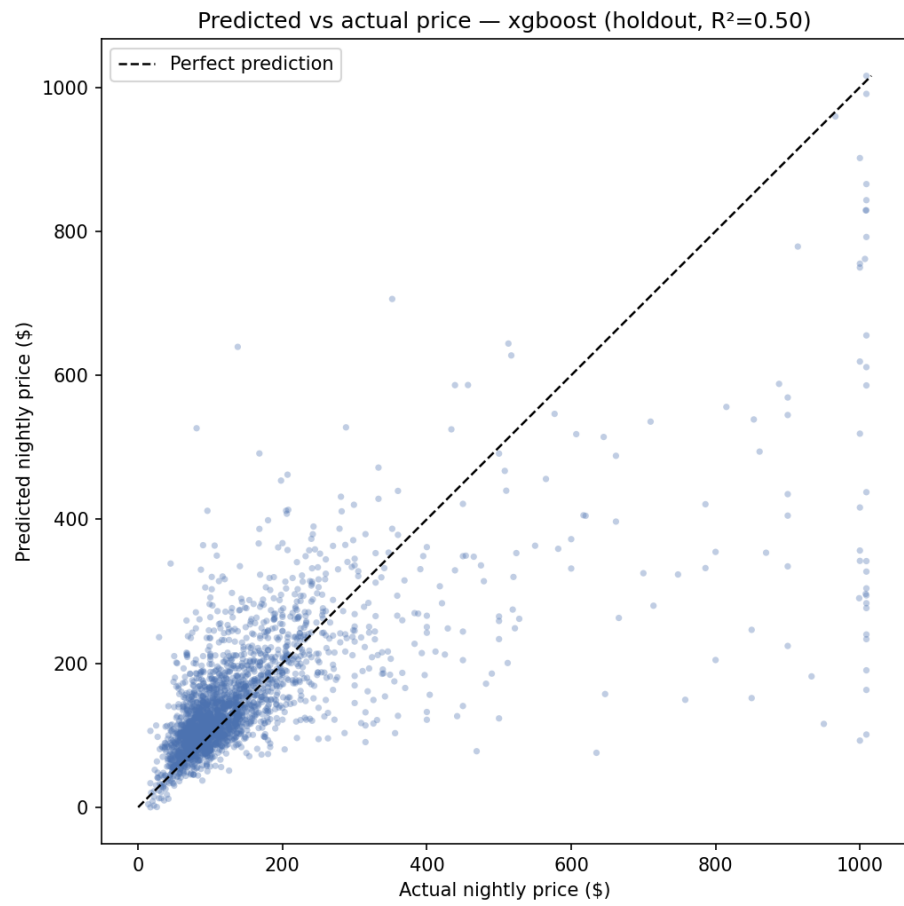
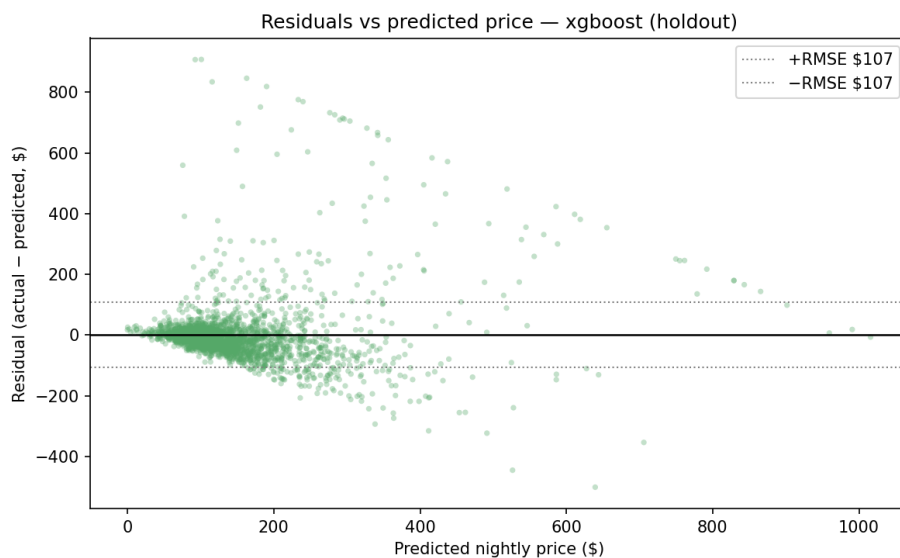


Figure 3: corr_heatmap



Points close to the dashed line are accurate predictions; points above it mean the model under-predicted.



Residual = actual - predicted. A cloud centered on the zero line means errors are balanced rather than systematically too high or too low.

Models & results

Model	Validation RMSE
Random Forest	\$121
XGBoost	\$112
TabPFN (capped sample, not directly comparable)	\$108

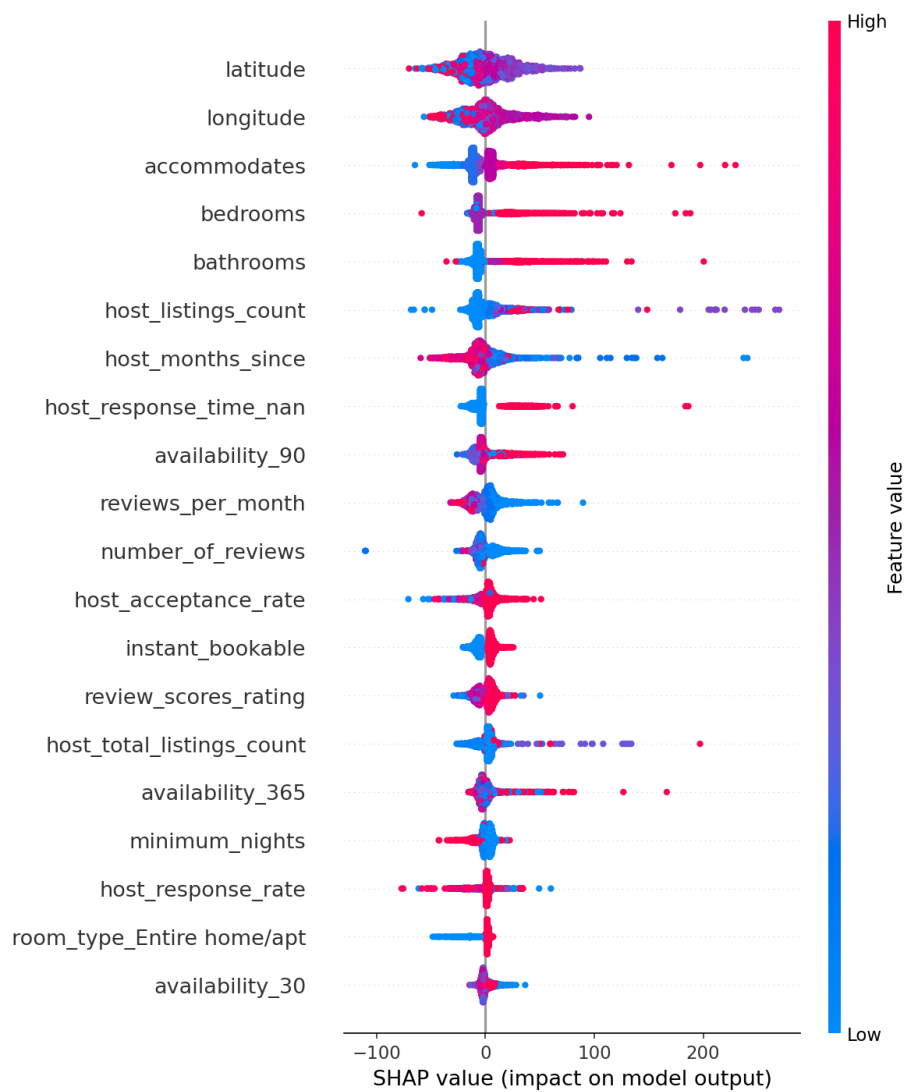
The model with the lowest validation RMSE was then scored once on the holdout set — see the accuracy section above for its final numbers.

Top features

The listing attributes that influence price the most (importance = how much the model relies on them):

Feature	Importance	Meaning
room_type_Entire home/apt	0.053	the room type (entire home vs private/shared room) (specifically Entire home/apt)

Feature	Importance	Meaning
room_type_Private room	0.052	the room type (entire home vs private/shared room) (specifically Private room)
bedrooms	0.045	the number of bedrooms
bathrooms	0.044	the number of bathrooms
instant_bookable	0.030	whether the listing is instantly bookable
host_response_time_nan	0.030	how quickly the host responds (specifically nan)
accommodates	0.026	how many guests a listing sleeps
host_neighbourhood_South Kensington	0.026	the host's neighbourhood (specifically South Kensington)
host_listings_count	0.025	how many listings the host manages
host_total_listings_count	0.025	a key listing attribute



SHAP summary: each dot is one listing; colour shows the feature's value (red = high, blue = low) and position shows whether it pushed the predicted price up (right) or down (left).

Prediction examples

A few real holdout listings with the model's prediction:

Listing	Actual	Predicted	Error	Error %
accommodates= \$ 155 bedrooms=1, room_type=Private room		\$160	\$5	3.5%
accommodates= \$ 127 bedrooms=2, room_type=Entire home/apt		\$151	\$24	18.7%
accommodates= \$ 78 bedrooms=1, room_type=Entire home/apt		\$124	\$46	59.6%
accommodates= \$ 97 bedrooms=1, room_type=Entire home/apt		\$137	\$40	41.1%
accommodates= \$ 1,009 bedrooms=1, room_type=Entire home/apt		\$1,016	\$7	0.7%
accommodates= \$ 76 bedrooms=1, room_type=Entire home/apt		\$87	\$11	14.9%
accommodates= \$ 131 bedrooms=1, room_type=Entire home/apt		\$135	\$4	2.9%
accommodates= \$ 89 bedrooms=1, room_type=Entire home/apt		\$88	\$1	1.0%

Recommendations

- The best-performing model was **XGBoost** (holdout RMSE \$107, MAE \$54, R^2 0.50).
- On validation, **XGBoost** beat Random Forest by 7% on RMSE (\$112 vs \$121). A simple ensemble or blending the two may squeeze out further gains.
- Pricing is most strongly driven by **room_type_Entire home/apt**, **room_type_Private room**, **bedrooms** — i.e. **room_type_Entire**

home/apt (the room type (entire home vs private/shared room) (specifically Entire home/apt)); room_type_Private room (the room type (entire home vs private/shared room) (specifically Private room)); bedrooms (the number of bedrooms). Hosts can use these levers when setting nightly rates.

- On average, predictions are off by about **38%** of the actual price (median error 25%), and **50%** of predictions landed within +/-25% of the true price.
- Accuracy caveat: the holdout RMSE of \$107 is **99% of the median listing price (\$108)**, so predictions for typical listings are expected to be off by roughly +/- \$107.
- Next steps: add richer features (neighbourhood_cleansed, amenities, review scores breakdowns), train on a log-transformed target to tame high-end outliers, run a deeper hyperparameter search, and extend the pipeline to more cities.

Limitations

- The model only “sees” the features listed in the config — it cannot capture neighbourhood charm, amenities, photos, or seasonality.
- Extreme high-end prices are clipped at the 99th percentile during cleaning, so predictions at the very top of the market are less meaningful.
- The accuracy numbers are averages: individual predictions can be much better or much worse.
- This report is generated automatically from the pipeline artifacts and reflects the data as of 2026-08-25.

*Automated pipeline: fetch → preprocess → EDA → train → evaluate → report.
Code in `src/`, config in `configs/base.yaml`.*